

CompTIA Security+ (SY0-701)

Course-End Project

Brute Force Attack Investigation, Response, and Mitigation on CentOS 7

Phase 0: Environment Setup and Initial Configuration

Objective:

Prepare a secure CentOS environment suitable for authentication testing and incident response.

Commands Executed:

```
useradd student
```

Result: Created a standard non-root user account.

```
usermod -aG wheel student
```

Result: Granted sudo privileges to the student user.

```
systemctl enable sshd  
systemctl start sshd  
systemctl status sshd
```

Result: SSH service enabled and running.

Evidence:

phase_0 screenshots document installation and SSH service status.

Phase 1: Brute Force Detection and Log Analysis

Objective:

Generate and identify brute-force authentication attempts using system logs.

Commands Executed:

```
ssh invalid@localhost
ssh admin@localhost
ssh student@localhost
ssh root@localhost
```

Result: Generated failed SSH authentication attempts.

```
grep "Failed password" /var/log/secure
```

Result: Displayed repeated authentication failures.

```
grep "Failed password" /var/log/secure | awk '{print $9}' | sort |
uniq -c
```

Result: Extracted and counted targeted usernames.

Evidence:

phase_1 screenshots show log analysis and username extraction.

Phase 2: Username Verification and Risk Assessment

Objective:

Determine whether brute-force attempts targeted valid system accounts.

Commands Executed:

```
cut -d: -f1 /etc/passwd
```

Result: Listed local user accounts.

```
id root
id student
id invalid
```

Result: Confirmed valid and non-existent users.

Evidence:

phase_2 screenshots document user verification steps.

Phase 3: Security Enhancements, Troubleshooting, and Mitigation

Objective:

Reduce exposure to brute-force attacks and harden authentication services.

Commands Executed:

```
nano /etc/ssh/sshd_config
```

Result: SSH port changed to 2222.

```
sshd -t
```

Result: Configuration syntax validated.

```
systemctl restart sshd
```

Result: Initial failure due to SELinux enforcement.

```
getenforce  
setenforce 0
```

Result: SELinux set to permissive temporarily.

```
systemctl restart sshd  
systemctl status sshd
```

Result: SSH restarted successfully on new port.

```
nano /etc/security/pwquality.conf
```

Result: Password complexity enforced.

```
yum install fail2ban
```

Result: Failed due to CentOS 7 EOL repositories.

Evidence:

phase_3 screenshots document configuration changes and troubleshooting.

Phase 4: Continuous Monitoring and Reporting

Objective:

Ensure ongoing detection of authentication anomalies.

Commands Executed:

```
tail -f /var/log/secure
```

Result: Real-time authentication monitoring.

```
grep "Failed password" /var/log/secure
```

Result: Historical review of failed login attempts.

Evidence:

phase_4 screenshots document monitoring commands.

Conclusion:

This project demonstrated a full incident response lifecycle for a brute-force attack under real-world constraints, emphasizing troubleshooting, layered defense, and security best practices aligned with CompTIA Security+ objectives.

As part of the security hardening strategy, the implementation of multifactor authentication (MFA) for SSH access was evaluated. In Linux environments, MFA is commonly implemented using PAM modules such as Google Authenticator or Duo Security to require a one-time password in addition to the user's credentials.

Due to the end-of-life status of CentOS 7 and associated repository limitations within the lab environment, secure installation of MFA-related PAM modules could not be reliably completed. Deploying unsupported authentication components on an EOL system may introduce additional security risks.

In a production environment, MFA would be implemented by integrating SSH with a supported PAM-based MFA solution, ensuring that all remote administrative access requires both a password and a time-based one-time token. This control significantly reduces the risk of successful brute-force attacks, even if credentials are compromised.